Trig Substitution Notes

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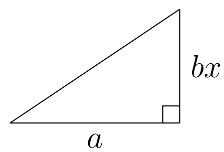
1 Introduction

What's the idea behind *Trigonometric Substitution*, or simply Trig Sub? What if we see an integral of the form

$$\int \frac{\sqrt{a^2 + b^2x^2}}{bx} \, \mathrm{d}x,$$

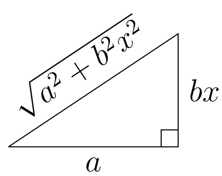
how would we go about solving this? The idea is that the square root is hiding a secret: there is a triangle hidden somewhere in this integral. Why would triangles have anything to do with integrals you might ask? Well that's the secret of trig sub! Let's look at a right triangle with side lengths a and bx:



What is the length of the hypotenuse of this triangle? Well, the Pythagorean Theorem tells us that the hypotenuse is

$$c^2 = a^2 + (bx)^2,$$
$$c = \sqrt{a^2 + b^2x^2}.$$

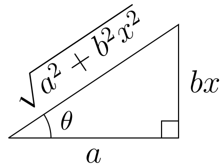
Therefore, our complete triangle is



By some miracle, we have found that our integrand is merely a ratio of side lengths of our triangle! (namely the hypotenuse divided by the right side length.) This is where our old friend from high school geometry comes in, trig functions. The 6 basic trig functions are defined as ratios of side lengths of a right triangle, which I have collected in the table below:

Function	Definition	Mnemonic
sin	$\frac{\text{opposite}}{\text{hypotenuse}}$	SOH
cos	$\frac{\text{adjacent}}{\text{hypotenuse}}$	CAH
tan	$\frac{\text{opposite}}{\text{adjacent}}$	TOA
csc	$\frac{\text{hypotenuse}}{\text{opposite}}$	$1/\sin$
sec	$\frac{\text{hypotenuse}}{\text{adjacent}}$	$1/\cos$
cot	$\frac{\text{adjacent}}{\text{opposite}}$	$1/\tan$

The terms "opposite", "adjacent", and "hypotenuse" refer to the sides of the triangle relative to the angle θ. If we place θ in our triangle from before,



then our corresponding trig functions are

$$\sin\theta = \frac{bx}{\sqrt{a^2+b^2x^2}}, \qquad \csc\theta = \frac{\sqrt{a^2+b^2x^2}}{bx},$$
$$\cos\theta = \frac{a}{\sqrt{a^2+b^2x^2}}, \qquad \sec\theta = \frac{\sqrt{a^2+b^2x^2}}{a},$$
$$\tan\theta = \frac{bx}{a}, \qquad \cot\theta = \frac{a}{bx}.$$

If we look at the integrand again, we can see that it is exactly $\csc\theta$! Substituting this into our integral, we get

$$\int \csc\theta \,\mathrm{d}x.$$

But we can't just substitute θ into our integral, we need to change the variable of integration from x to θ. This is where the chain rule comes in. We can use the chain rule to find the derivative of x with respect to θ, and then substitute that into our integral. If we look back at our trig functions, we can see that

$$\tan\theta = \frac{bx}{a}$$

Therefore, we can solve for x in terms of θ:

$$x = \frac{a}{b}\tan\theta.$$

Now we can differentiate this with respect to θ:

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{a}{b}\sec^2\theta.$$

This means that

$$\mathrm{d}x = \frac{a}{b}\sec^2\theta \,\mathrm{d}\theta.$$

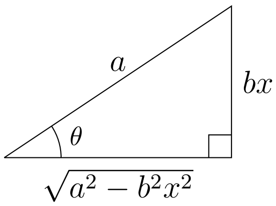
Now we can substitute this into our integral:

$$\int \csc\theta \,\mathrm{d}x = \int \csc\theta \cdot \frac{a}{b}\sec^2\theta \,\mathrm{d}\theta.$$

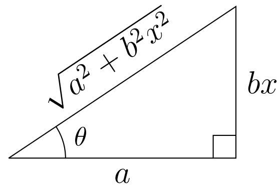
This in principle is a much simpler integral to solve using our knowledge of trig identities and integrals.

I will end by make note of an important fact about trig sub: there are only three types of trig substitutions:

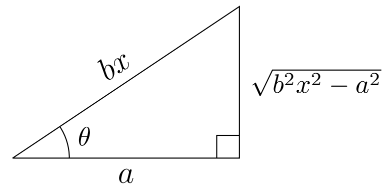
$x = \frac{a}{b}\sin\theta$



$x = \frac{a}{b}\tan\theta$



$x = \frac{a}{b}\sec\theta$



The way we determine which substitution to use is by looking at the integrand. If there is a plus sign in the square root, we use the second substitution. If there is a minus sign in the square root, and x is on the left, we use the last substitution. If there is a minus sign in the square root, and x is on the right, we use the first substitution. There are technically 3 other ways to set up the substitutions, but they are all equivalent to the ones above. The reason we use these ones is because they minimize the minus signs floating around as a result of the substitution.

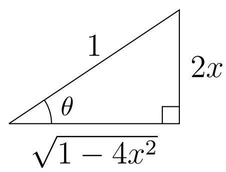
2 Examples

2.1 Example 1

Evaluate

$$\int \sqrt{1-4x^2}\,\mathrm{d}x$$

Solution. First, we need to identify the type of trig sub we will be using. We see that there is a minus sign in the square root, and x is on the right, so we will use the first substitution, or $\sin\theta$ substitution. Next, we will set up our triangle:



(Note that there is a 2 in front of x because we want $(2x)^2$ to be $4x^2$.) Next, we will write down our trig functions:

$$\begin{aligned}
\sin\theta &= \frac{2x}{1} = 2x, & \csc\theta &= \frac{1}{2x},\\
\cos\theta &= \frac{\sqrt{1-4x^2}}{1} = \sqrt{1-4x^2}, & \sec\theta &= \frac{1}{\sqrt{1-4x^2}},\\
\tan\theta &= \frac{2x}{\sqrt{1-4x^2}}, & \cot\theta &= \frac{\sqrt{1-4x^2}}{2x}.
\end{aligned}$$

Now we can plug these into our integral:

$$\int \sqrt{1-4x^2}\,\mathrm{d}x = \int \cos\theta\,\mathrm{d}x.$$

However, we need to change the variable of integration from x to θ. We can do this by using the chain rule. We can use the fact that

$$\begin{gathered}
\sin\theta = 2x\\
\frac{\mathrm{d}}{\mathrm{d}\theta}\sin\theta = 2\frac{\mathrm{d}x}{\mathrm{d}\theta}\\
\mathrm{d}x = \frac{1}{2}\cos\theta\,\mathrm{d}\theta.
\end{gathered}$$

Now we can substitute this into our integral:

$$\int \cos\theta\,\mathrm{d}x = \int \cos\theta\cdot\frac{1}{2}\cos\theta\,\mathrm{d}\theta = \frac{1}{2}\int \cos^2\theta\,\mathrm{d}\theta.$$

Now we can use the double angle formula to reduce the power of cosine[1]:

$$\frac{1}{2}\int \cos^2\theta\,\mathrm{d}\theta = \frac{1}{2}\int \frac{1+\cos(2\theta)}{2}\,\mathrm{d}\theta = \frac{1}{4}\int 1+\cos(2\theta)\,\mathrm{d}\theta.$$

[1] If you don't remember these, they are in the Useful Identities and Formulas section.

Now we can integrate term by term:

$$\frac{1}{4}\int 1+\cos(2\theta)\,\mathrm{d}\theta = \frac{1}{4}\left(\int 1\,\mathrm{d}\theta + \int \cos(2\theta)\,\mathrm{d}\theta\right) = \frac{1}{4}\left(\theta + \frac{1}{2}\sin(2\theta)\right) + C.$$

Simplifying this, we get

$$\frac{1}{4}\left(\theta + \frac{1}{2}\sin(2\theta)\right) + C = \frac{1}{4}\theta + \frac{1}{8}\sin(2\theta) + C.$$

Now we need to substitute back in for θ. The first term is easy, if we look back at our triangle, we can see that

$$\theta = \arcsin(2x).$$

The second term is a bit trickier, but we can use the double angle formula to rewrite it as

$$\sin(2\theta) = 2\sin\theta\cos\theta$$

Now we can substitute back in for $\sin\theta$ and $\cos\theta$:

$$\sin(2\theta) = 2\sin\theta\cos\theta = 2(2x)\sqrt{1-4x^2} = 4x\sqrt{1-4x^2}.$$

Putting this all together, we get

$$\boxed{\int \sqrt{1-4x^2}\,\mathrm{d}x = \frac{1}{4}\arcsin(2x) + \frac{1}{2}x\sqrt{1-4x^2} + C}.$$

2.2 Example 2

Evaluate
$$\int \frac{1}{x^2+2x+2}\,\mathrm{d}x.$$

Solution. Although this may not initially look like a trig sub, if we do some algebra, we can see that it is. First, we can complete the square:

$$x^2+2x+2 = x^2+2x+1+1 = (x+1)^2+1.$$

Putting this back into our integral, we get

$$\int \frac{1}{(x+1)^2+1}\,\mathrm{d}x.$$

With a quick u-sub, we can see that this is a trig sub integral (using $u = x+1$):

$$\int \frac{1}{u^2+1}\,\mathrm{d}u$$

Although there are no square roots in this integral, if we look at the denominator, we can see that it is a sum of squares, or
$$\int \frac{1}{\left(\sqrt{u^2+1}\right)^2}\,\mathrm{d}u.$$

This means that we can use the $\tan\theta$ substitution:

$$u = \tan\theta$$
$$\frac{\mathrm{d}u}{\mathrm{d}\theta} = \frac{\mathrm{d}}{\mathrm{d}\theta}\tan\theta = \sec^2\theta$$
$$\mathrm{d}u = \sec^2\theta\,\mathrm{d}\theta.$$

Now we can substitute this into our integral:

$$\int \frac{1}{\tan^2\theta+1}\cdot\sec^2\theta\,\mathrm{d}\theta.$$

Now we can use the Pythagorean identity to rewrite the denominator:

$$\int \frac{1}{\sec^2\theta}\cdot\sec^2\theta\,\mathrm{d}\theta = \int 1\,\mathrm{d}\theta.$$

Now we can integrate this to get
$$\int 1\,\mathrm{d}\theta = \theta + C.$$

Now we need to substitute back in for θ. If we look back at our substitution, we can see that

$$\theta = \arctan(u).$$

Now we can substitute back in for u:

$$\theta = \arctan(x+1).$$

Putting this all together, we get

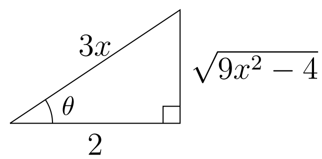
$$\boxed{\int \frac{1}{x^2+2x+2}\,\mathrm{d}x = \arctan(x+1) + C}.$$

2.3 Example 3

Evaluate

$$\int_{\frac{4}{3}}^{\infty} \frac{1}{x^2\sqrt{9x^2-4}}\,\mathrm{d}x.$$

Solution. First, we need to identify the type of trig sub we will be using. We see that there is a minus sign in the square root, and x is on the left, so we will use the $\sec\theta$ substitution. Next, we will set up our triangle:



Next, we will write down our trig functions:

$$\begin{aligned}
\sin\theta &= \frac{\sqrt{9x^2-4}}{3x}, & \csc\theta &= \frac{3x}{\sqrt{9x^2-4}},\\
\cos\theta &= \frac{2}{3x}, & \sec\theta &= \frac{3x}{2},\\
\tan\theta &= \frac{\sqrt{9x^2-4}}{2}, & \cot\theta &= \frac{2}{\sqrt{9x^2-4}}.
\end{aligned}$$

We can see that

$$\frac{1}{x^2} = \frac{3^2}{2^2}\cdot\frac{2}{3x}\cdot\frac{2}{3x} = \frac{9}{4}\cos^2\theta.$$

Similarly, we can see that

$$\frac{1}{\sqrt{9x^2-4}} = \frac{1}{2}\cdot\frac{2}{\sqrt{9x^2-4}} = \frac{1}{2}\cot\theta.$$

Plugging these into our integral, we get

$$\int_{\frac{4}{3}}^{\infty} \frac{1}{x^2\sqrt{9x^2-4}}\,\mathrm{d}x = \int_{x=\frac{4}{3}}^{x=\infty} \frac{9}{4}\cos^2\theta\cdot\frac{1}{2}\cot\theta\,\mathrm{d}x.$$

Now we need to change the variable of integration from x to θ. We can use

$$\frac{3}{2}x = \sec\theta$$

$$\frac{3}{2}\frac{\mathrm{d}x}{\mathrm{d}\theta} = \sec\theta\tan\theta$$

$$\mathrm{d}x = \frac{2}{3}\sec\theta\tan\theta\,\mathrm{d}\theta.$$

Now we can substitute this into our integral:

$$\int_{x=\frac{4}{3}}^{x=\infty} \frac{9}{4}\cos^2\theta\cdot\frac{1}{2}\cot\theta\cdot\frac{2}{3}\sec\theta\tan\theta\,\mathrm{d}\theta.$$

Simplifying this, we get

$$\frac{3}{4}\int_{x=\frac{4}{3}}^{x=\infty} \cos\theta \, \mathrm{d}\theta.$$

Now, we need to change the limits of integration. If we look back at our substitution, we can see that

$$\frac{3}{2}x = \sec\theta$$

For the lower limit, we have

$$\begin{aligned}\frac{3}{2}\cdot\frac{4}{3} = 2 &= \sec\theta \\ \frac{1}{2} &= \cos\theta \\ \theta &= \frac{\pi}{3}.\end{aligned}$$

Similarly, for the upper limit, we have

$$\begin{aligned}\frac{3}{2}\cdot\infty = \infty &= \sec\theta \\ \frac{1}{\infty} = 0 &= \cos\theta \\ \theta &= \frac{\pi}{2}.\end{aligned}$$

Putting this all together, we get

$$\frac{3}{4}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos\theta \, \mathrm{d}\theta.$$

Now we can integrate this to get

$$\frac{3}{4}\left(\sin(\theta)\Big|_{\frac{\pi}{3}}^{\frac{\pi}{2}}\right) = \frac{3}{4}\left(\sin\frac{\pi}{2} - \sin\frac{\pi}{3}\right) = \frac{3}{4}\left(1 - \frac{\sqrt{3}}{2}\right).$$

Therefore, the final answer is

$$\boxed{\int_{\frac{4}{3}}^{\infty} \frac{1}{x^2\sqrt{9x^2-4}}\,\mathrm{d}x = \frac{3}{4}\left(1 - \frac{\sqrt{3}}{2}\right)}.$$

3 Problems

Rewrite the following using a trig substitution.

1. $\sqrt{4-x^2}$
2. $\sqrt{13+25x^2}$
3. $(7x^2-3)^{\frac{5}{2}}$
4. $\sqrt{(u+3)^2-100}$
5. $\sqrt{4(9t-5)^2+1}$
6. $\sqrt{1-4z-2z^2}$
7. $(x^2-8x+21)^{\frac{3}{2}}$
8. $\sqrt{e^{8x}-9}$

For the following integrals, use a trig substitution to evaluate them.

9. $\int \frac{\sqrt{x^2+16}}{x^4}\,dx$
10. $\int \sqrt{1-7w^2}\,dw$
11. $\int t^3\left(3t^2-4\right)^{\frac{5}{2}}\,dt$
12. $\int \frac{1}{\sqrt{9x^2-36x+37}}\,dx$
13. $\int \frac{(z+3)^5}{(40-6z-z^2)^{\frac{3}{2}}}\,dz$
14. $\int \cos(x)\sqrt{9+25\sin^2(x)}\,dx$

Solutions

1.
$$\sqrt{4-x^2}=\sqrt{4}\cdot\sqrt{1-\frac{1}{4}x^2}=2\cdot\cos\theta.$$

2.
$$\sqrt{13+25x^2}=\sqrt{13}\cdot\sqrt{1+\frac{25}{13}x^2}=\sqrt{13}\cdot\sec\theta.$$

3.
$$\left(7x^2-3\right)^{\frac{5}{2}}=(3)^{\frac{5}{2}}\cdot\left(\frac{7}{3}x^2-1\right)^{\frac{5}{2}}=(3)^{\frac{5}{2}}\cdot(\tan\theta)^{\frac{5}{2}}.$$

4.
$$\sqrt{(u+3)^2-100}=\sqrt{100}\cdot\sqrt{\left(\frac{u+3}{10}\right)^2-1}=10\cdot\tan\theta.$$

5.
$$\sqrt{4(9t-5)^2+1}=\sqrt{\tan^2\theta+1}=\sec\theta.$$

6.
$$\sqrt{1-4z-2z^2}=\sqrt{3-2\left(z+1\right)^2}=\sqrt{3}\cdot\sqrt{1-\frac{2}{3}\left(z+1\right)^2}=\sqrt{3}\cdot\cos\theta.$$

7.
$$\left(x^2-8x+21\right)^{\frac{3}{2}}=\left((x-4)^2+5\right)^{\frac{3}{2}}=(5)^{3/2}\cdot\left(\frac{(x-4)^2}{5}+1\right)^{\frac{3}{2}}=(5)^{3/2}\cdot\sec^3\theta.$$

8.
$$\sqrt{e^{8x}-9}=\sqrt{9}\cdot\sqrt{\frac{e^{8x}}{9}-1}=3\cdot\sqrt{\left(\frac{e^{4x}}{3}\right)^2-1}=3\cdot\tan\theta.$$

9.
$$\int\frac{\sqrt{x^2+16}}{x^4}\,dx=\frac{1}{16}\int\frac{\cos\theta}{\sin^4\theta}\,d\theta=-\frac{\left(x^2+16\right)^{3/2}}{48x^3}+C.$$

10.
$$\int\sqrt{1-7w^2}\,dw=\frac{1}{\sqrt{7}}\int\cos^2\theta\,d\theta=\frac{1}{2\sqrt{7}}\left[\sin^{-1}\left(\sqrt{7}w\right)+\sqrt{7}w\sqrt{1-7w^2}\right]+C.$$

11.
$$\int t^3\left(3t^2-4\right)^{\frac{5}{2}}\,dt=\frac{512}{9}\int\sec^4\theta\tan^6\theta\,d\theta=\frac{\left(3t^2-4\right)^{\frac{9}{2}}}{81}+\frac{4\left(3t^2-4\right)^{\frac{7}{2}}}{63}+C.$$

Solutions continued

12.

$$\begin{aligned}\int \frac{1}{\sqrt{9x^2-36x+37}}\,dx &= \int \frac{1}{\sqrt{9\left(x-2\right)^2+1}} \\ &= \frac{1}{3}\int \sec\theta\,d\theta \\ &= \frac{1}{3}\ln\left|\sqrt{9\left(x-2\right)^2+1}+3\left(x-2\right)\right|+C.\end{aligned}$$

13.

$$\begin{aligned}\int \frac{(z+3)^5}{(40-6z-z^2)^{\frac{3}{2}}}\,dz &= \int \frac{\left(z+3\right)^5}{\left(49-\left(z+3\right)^2\right)^{\frac{3}{2}}}\,dz \\ &= 343\int \frac{\sin^5\theta}{\cos^2\theta}\,d\theta \\ &= \frac{2401}{\sqrt{49-\left(z+3\right)^2}}+98\sqrt{49-\left(z+3\right)^2}-\frac{\left(49-\left(z+3\right)^2\right)^{3/2}}{3}+C.\end{aligned}$$

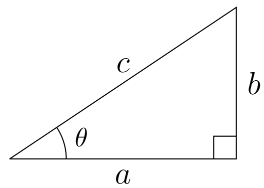
14.

$$\begin{aligned}\int \cos(x)\sqrt{9+25\sin^2(x)}\,dx &= \int \sqrt{9+25u^2}\,du \\ &= \frac{9}{5}\int \sec^3\theta\,d\theta \\ &= \frac{9}{5}\left[\sec\theta\tan\theta+\ln\left|\sec\theta+\tan\theta\right|\right]+C \\ &= \frac{u\sqrt{9+25u^2}}{2}+\frac{9}{10}\ln\left|\frac{u}{3}+\sqrt{\frac{9}{25}+u^2}\right|+C.\end{aligned}$$

4 Useful Identities and Formulas

4.1 Basics

If we have a right triangle with side lengths a, b, and c, and angle θ, then we have



$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{b}{c}, \quad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{a}{c}.$$

From only these two trig functions, we can actually derive the remaining four! We know that

$$\tan\theta = \frac{b}{a}$$

However, we can do the clever trick of "multiplying by 1" to get

$$\tan\theta = \frac{b}{a} \cdot \frac{\frac{1}{c}}{\frac{1}{c}} = \frac{b/c}{a/c} = \frac{\sin\theta}{\cos\theta}.$$

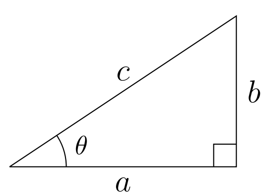
The remaining three functions are exactly as the reciprocals of the first three, so they are immediately

$$\csc\theta = \frac{1}{\sin\theta}, \quad \sec\theta = \frac{1}{\cos\theta}, \quad \cot\theta = \frac{1}{\tan\theta}.$$

Incredibly, we have recovered all 6 trig functions from only two of them![2]

4.2 Pythagorean Identities

The titular Pythagorean identity comes from the Pythagorean Theorem. If we start with our standard right triangle, then we have



The Pythagorean Theorem tells us that

$$a^2 + b^2 = c^2.$$

[2] In practice, this can be done with almost any two of the six (you cannot choose both a trig function and its reciprocal), but sin and cos are the most common.

If we divide both sides by c^2, we get

$$\frac{a^2}{c^2} + \frac{b^2}{c^2} = 1.$$

Let's now think about things in terms of θ. From our definitions of the trig functions, we have

$$\sin\theta = \frac{b}{c}, \quad \cos\theta = \frac{a}{c}.$$

We can substitute these into our equation to get

$$\cos^2\theta + \sin^2\theta = 1.$$

This is the most famous of the Pythagorean identities, often called *the* Pythagorean identity. However, there are two other Pythagorean identities that are derived from this one. For the first, we can divide both sides by $\cos^2\theta$:

$$1 + \frac{\sin^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta}.$$

We can simply this using our basic trig identities:

$$1 + \tan^2\theta = \sec^2\theta,$$

which is the second Pythagorean identity. For the third (and least common) Pythagorean identity, we can divide both sides by $\sin^2\theta$:

$$\frac{\cos^2\theta}{\sin^2\theta} + 1 = \frac{1}{\sin^2\theta}.$$

Again, we can simplify this using our basic trig identities:

$$\cot^2\theta + 1 = \csc^2\theta.$$

4.3 Angle Sum and Difference Formulas

The derivations of these formulas are quite complex and a bit contrived, so I will not include them here. However, they are very useful and worth memorizing.

$$\sin(\theta+\phi) = \sin\theta\cos\phi + \cos\theta\sin\phi, \qquad \sin(\theta-\phi) = \sin\theta\cos\phi - \cos\theta\sin\phi,$$
$$\cos(\theta+\phi) = \cos\theta\cos\phi - \sin\theta\sin\phi, \qquad \cos(\theta-\phi) = \cos\theta\cos\phi + \sin\theta\sin\phi.$$

A useful mnemonic I like to use for them is

"Sine same sign, cosine same trig function."

Admittedly, this is a bit of a stretch, but if we look at the formulas, we can see that the sine functions have the same sign as the angle they are being added to or subtracted from while having different trig functions, while the cosine functions have the same trig function but opposite signs as the angle they are being added to or subtracted from.

Two very useful corollaries of these formulas are the double angle formulas:

$$\sin(2\theta) = 2\sin\theta\cos\theta, \qquad \cos(2\theta) = \cos^2\theta - \sin^2\theta.$$

In terms of trig sub, the most useful of these is the cosine double angle formula. If we examine two alternate forms of this formula:

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta = \cos^2\theta - (1-\cos^2\theta) = 2\cos^2\theta - 1.$$

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta = (1-\sin^2\theta) - \sin^2\theta = 1 - 2\sin^2\theta,$$

we can see that with some algebra, we can express $\sin^2\theta$ and $\cos^2\theta$ in terms of $\cos(2\theta)$. This now gives us a way to integrate powers of sine and cosine!

4.4 Basic Calculus of Trig

The two fundamental trig derivatives are

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin x = \cos x, \quad \frac{\mathrm{d}}{\mathrm{d}x}\cos x = -\sin x.$$

There's not a very elegant derivation of these, but they are still very useful to know. The remaining four basic trig derivatives can be derived from these two:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\tan x &= \frac{\mathrm{d}}{\mathrm{d}x}\frac{\sin x}{\cos x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x,\\
\frac{\mathrm{d}}{\mathrm{d}x}\cot x &= \frac{\mathrm{d}}{\mathrm{dx}}\frac{1}{\tan x} = \frac{0\cdot\tan^2 x - 1\cdot\sec^2 x}{\tan^4 x} = -\frac{\sec^2 x}{\tan^4 x} = -\csc^2 x,\\
\frac{\mathrm{d}}{\mathrm{d}x}\sec x &= \frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{\cos x} = \frac{\sin x}{\cos^2 x} = \sec x\tan x,\\
\frac{\mathrm{d}}{\mathrm{d}x}\csc x &= \frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{\sin x} = -\frac{\cos x}{\sin^2 x} = -\csc x\cot x.
\end{aligned}$$

As for the integrals, there are two slightly different categories of them. The first category is what I like to call the "anti-derivatives." Although all integrals are technically anti-derivatives, these are the ones that are the most straightforward to derive as they are the anti-derivatives of the derivatives we just found.

$$\begin{aligned}
&\int \sin x\,\mathrm{d}x = -\cos x + C, && \int \csc x\cot x\,\mathrm{d}x = -\csc x + C,\\
&\int \cos x\,\mathrm{d}x = \sin x + C, && \int \sec x\tan x\,\mathrm{d}x = \sec x + C,\\
&\int \sec^2 x\,\mathrm{d}x = \tan x + C, && \int \csc^2 x\,\mathrm{d}x = -\cot x + C.
\end{aligned}$$

The second category is what I like to call the "trig integrals." These are the integrals that are not as straightforward, but rather tell us what the integrals of the trig functions are. We already know the integrals of $\sin x$ and $\cos x$ from the anti-derivatives, but the remaining four are a bit more complicated. I will provide the derivations for the first two, and the remaining two are left as an exercise. The first we will be deriving is the integral of $\tan x$.

$$\int \tan x\,\mathrm{d}x = \int \frac{\sin x}{\cos x}\,\mathrm{d}x.$$

We can use the substitution $u = \cos x$, which gives us $\mathrm{d}u = -\sin x \,\mathrm{d}x$. This means that

$$\int \tan x \,\mathrm{d}x = -\int \frac{1}{u} \,\mathrm{d}u.$$

We can now integrate this to get

$$\int \tan x \,\mathrm{d}x = -\ln|u| + C.$$

Substituting back in for u, we get

$$\int \tan x \,\mathrm{d}x = -\ln|\cos x| + C.$$

The second integral we will be deriving is the integral of $\sec x$.

$$\int \sec x \,\mathrm{d}x.$$

We now fall back on the classic mathematician's stratagem of "multiplying by 1" to get

$$\int \sec x \,\mathrm{d}x = \int \frac{\sec x(\sec x + \tan x)}{\sec x + \tan x} \,\mathrm{d}x.$$

With this clever trick, we can now use the substitution $u = \sec x + \tan x$, which gives us $\mathrm{d}u = (\sec x \tan x + \sec^2 x) \,\mathrm{d}x$. This means that

$$\int \sec x \,\mathrm{d}x = \int \frac{1}{u} \,\mathrm{d}u.$$

We can now integrate this to get

$$\int \sec x \,\mathrm{d}x = \ln|u| + C.$$

Substituting back in for u, we get

$$\int \sec x \,\mathrm{d}x = \ln|\sec x + \tan x| + C.$$

The remaining two integrals are left as an exercise for the reader. They are

$$\int \csc x \,\mathrm{d}x = -\ln|\csc x + \cot x| + C, \quad \int \cot x \,\mathrm{d}x = \ln|\sin x| + C.$$

4.5 Integrals of Powers of Trig Functions

The types of integrals we will be looking at here are

$$\int \sin^n \cos^m x \,\mathrm{d}x,$$

where n and m are non-negative integers. The general form can be quite daunting, so we will instead examine some useful cases. The first case we will look at is when $n = 1$ and $m = 1$:

$$\int \sin x \cos x \, \mathrm{d}x.$$

If we let $u = \sin x$, then $\mathrm{d}u = \cos x \, \mathrm{d}x$. This means that

$$\int \sin x \cos x \, \mathrm{d}x = \int u \, \mathrm{d}u = \frac{u^2}{2} + C = \frac{\sin^2 x}{2} + C.$$

Let's now look at the case when $n = 2$ and $m = 1$:

$$\int \sin^2 x \cos x \, \mathrm{d}x.$$

Once again, we can use the substitution $u = \sin x$, which gives us $\mathrm{d}u = \cos x \, \mathrm{d}x$. This means that

$$\int \sin^2 x \cos x \, \mathrm{d}x = \int u^2 \, \mathrm{d}u = \frac{u^3}{3} + C = \frac{\sin^3 x}{3} + C.$$

Following this pattern, we can see that in general, if $m = 1$, then

$$\int \sin^n x \cos x \, \mathrm{d}x = \frac{\sin^{n+1} x}{n+1} + C.$$

Similarly for cosine, $n = 1$, then

$$\int \sin x \cos^m x \, \mathrm{d}x = -\frac{\cos^{m+1} x}{m+1} + C,$$

where the negative sign comes from the fact that $\mathrm{d}u = -\sin x \, \mathrm{d}x$. The next case we will look at is when m is any odd number:

$$\int \sin^n x \cos^m x \, \mathrm{d}x.$$

First, we will peel off a single cosine term:

$$\int \sin^n x \cos^m x \, \mathrm{d}x = \int \sin^n x \cos^{m-1} x \cos x \, \mathrm{d}x.$$

(Note that as we are assuming m is odd, it is always possible to peel off a single cosine term and have the remaining cosine term be even.) Now, as the remaining cosine term is even, we can use the Pythagorean identity to rewrite it as

$$\int \sin^n x \cos^{m-1} x \cos x \, \mathrm{d}x = \int \sin^n x (1 - \sin^2 x)^{\frac{m-1}{2}} \cos x \, \mathrm{d}x.$$

Now we can use the substitution $u = \sin x$, which gives us $\mathrm{d}u = \cos x \, \mathrm{d}x$. This means that

$$\int \sin^n x (1 - \sin^2 x)^{\frac{m-1}{2}} \cos x \, \mathrm{d}x = \int u^n (1 - u^2)^{\frac{m-1}{2}} \, \mathrm{d}u.$$

Although in this general form this is rather cumbersome, what is important to note is that we have reduced the integral into an integral of a polynomial, which is much easier to solve. The case for odd sine is similar, but with a negative sign:

$$\int \sin^m x \cos^n x \, \mathrm{d}x = -\int u^m (1-u^2)^{\frac{n-1}{2}} \, \mathrm{d}u.$$

The final case we will look at is when both n and m are even:

$$\int \sin^n x \cos^m x \, \mathrm{d}x.$$

In this case, we again use the Pythagorean identity to rewrite the integral as

$$\int \sin^n x \cos^m x \, \mathrm{d}x = \int \sin^n x (1-\sin^2 x)^{\frac{m}{2}} \, \mathrm{d}x.$$

However, instead of having a regular polynomial as before, we now have a polynomial in sine. This means that we will have to expand the polynomial, use the double angle formula to reduce the powers of sine, and then integrate term by term. This is a bit more complicated than the previous cases, but I will show an example to illustrate the process. Let's look at the case when $n = 2$ and $m = 2$:

$$\int \sin^2 x \cos^2 x \, \mathrm{d}x.$$

We can use the Pythagorean identity to rewrite the integral as

$$\int \sin^2 x (1-\sin^2 x) \, \mathrm{d}x = \int \sin^2 x - \sin^4 x \, \mathrm{d}x.$$

Now we can use the double angle formula to reduce the powers of sine:

$$\int \sin^2 x - \sin^4 x \, \mathrm{d}x = \int \frac{1-\cos(2x)}{2} - \frac{(1-\cos(2x))^2}{4} \, \mathrm{d}x.$$

Now we can expand the polynomial:

$$\int \frac{1-\cos(2x)}{2} - \frac{1-2\cos(2x)+\cos^2(2x)}{4} \, \mathrm{d}x = \int \frac{1}{4} - \frac{\cos^2(2x)}{4} \, \mathrm{d}x.$$

Now we can use the double angle formula again to reduce the power of cosine:

$$\int \frac{1}{4} - \frac{1+\cos(4x)}{8} \, \mathrm{d}x = \int \frac{1}{8} - \frac{\cos(4x)}{8} \, \mathrm{d}x.$$

Now we can integrate term by term:

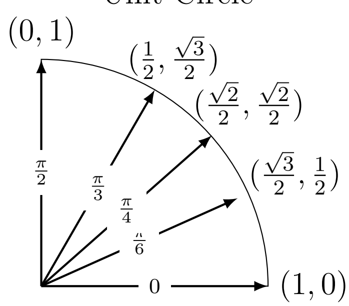
$$\int \frac{1}{8} - \frac{\cos(4x)}{8} \, \mathrm{d}x = \frac{x}{8} - \frac{\sin(4x)}{32} + C.$$

Formula Sheet

Function	Definition
sin	$\frac{\text{opposite}}{\text{hypotenuse}}$
cos	$\frac{\text{adjacent}}{\text{hypotenuse}}$
tan	$\frac{\text{opposite}}{\text{adjacent}}$
csc	$\frac{\text{hypotenuse}}{\text{opposite}}$
sec	$\frac{\text{hypotenuse}}{\text{adjacent}}$
cot	$\frac{\text{adjacent}}{\text{opposite}}$

Identity	Formula
Pythagorean (Standard)	$\sin^2\theta + \cos^2\theta = 1$
Pythagorean (Secant)	$1 + \tan^2\theta = \sec^2\theta$
Pythagorean (Cosecant)	$1 + \cot^2\theta = \csc^2\theta$
Angle Sum	$\sin(\theta+\phi) = \sin\theta\cos\phi + \cos\theta\sin\phi$
Angle Difference	$\sin(\theta-\phi) = \sin\theta\cos\phi - \cos\theta\sin\phi$
Cosine Sum	$\cos(\theta+\phi) = \cos\theta\cos\phi - \sin\theta\sin\phi$
Cosine Difference	$\cos(\theta-\phi) = \cos\theta\cos\phi + \sin\theta\sin\phi$
Double Angle (Sine)	$\sin(2\theta) = 2\sin\theta\cos\theta$
Double Angle (Cosine)	$\cos(2\theta) = \cos^2\theta - \sin^2\theta$
	$\cos(2\theta) = 2\cos^2\theta - 1$
	$\cos(2\theta) = 1 - 2\sin^2\theta$

Unit Circle



Integration and Differentiation

Derivative	Formula
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\csc^2 x$
$\sec x$	$\sec x\tan x$
$\csc x$	$-\csc x\cot x$

Integral	Formula
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$
$\tan x$	$-\ln\lvert\cos x\rvert + C$
$\cot x$	$\ln\lvert\sin x\rvert + C$
$\sec x$	$\ln\lvert\sec x + \tan x\rvert + C$
$\csc x$	$-\ln\lvert\csc x + \cot x\rvert + C$

Integral Power Reduction Formulas[3]

Power	Formula
$\int \sin^n x\,\mathrm{d}x$	$-\frac{\cos x\sin^{n-1}x}{n} + \frac{n-1}{n}\int \sin^{n-2}x\,\mathrm{d}x$
$\int \cos^n x\,\mathrm{d}x$	$\frac{\sin x\cos^{n-1}x}{n} + \frac{n-1}{n}\int \cos^{n-2}x\,\mathrm{d}x$
$\int \tan^n x\,\mathrm{d}x$	$\frac{\tan^{n-1}x}{n-1} + \frac{1}{n-1}\int \tan^{n-2}x\,\mathrm{d}x$
$\int \csc^n x\,\mathrm{d}x$	$-\frac{\csc^{n-2}x\cot x}{n-1} + \frac{n-2}{n-1}\int \csc^{n-2}x\,\mathrm{d}x$
$\int \sec^n x\,\mathrm{d}x$	$\frac{\sec^{n-2}x\tan x}{n-1} + \frac{n-2}{n-1}\int \sec^{n-2}x\,\mathrm{d}x$
$\int \cot^n x\,\mathrm{d}x$	$-\frac{\cot^{n-1}x}{n-1} + \frac{1}{n-1}\int \cot^{n-2}x\,\mathrm{d}x$

[3] I did not include the derivations for these as they are quite long and tedious, and often not very useful. I will, however, include these formulas here for the sake of completeness.